COHOMOLOGY OF A MODULE SHEAF ON THE FPPF SITE AGREES WITH THE COHOMOLOGY OF ITS SMALL-SITE RESTRICTION

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans carry 2 navigation stubs; the construction of the comparison map and the parasiticity argument are both inside the span, so nothing load bearing is delegated.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Editorial note. The label lemma-quasi-coherent-and-flat-base-change is defined only in descent.tex, and its canonical tag is not among the tags of the published items of this file, so it is not a published duplicate.

Allowed standard prerequisites (editorial).

- Parasitic sheaves for the fppf topology have vanishing cohomology. Same chapter, Lemma lemma-cohomology-parasitic, invoked at proof lines 4089-4092. Supplies the passage from the vanishing of the kernel and cokernel to the isomorphism on cohomology.
- Cohomology of a quasi-coherent module agrees on the small and big sites. Same chapter, Proposition proposition-same-cohomology-quasi-coherent, invoked at proof lines 4093-4095. Supplies the result for the pulled back quasi-coherent sheaf.
- Pullback of a sheaf on the small site to the big fppf site and the induced comparison map. Standard descent and site theory, used at proof lines 4073-4079. Supplies the sheaf F and the comparison map whose kernel and cokernel are examined.

Source reference. Original lemma 0: sources/stacks-chow/tags.txt, source lines 1–1. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original proposition 0: sources/stacks-chow/tags.txt, source lines 1–1. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let $\tau \in \{Zar, étale\}$. Let $\mathcal{G}$ be a sheaf of $\mathcal{O}$-modules on $(Sch/S)_{fppf}$ such that*

(1) *$\mathcal{G}|_{S_\tau}$ is quasi-coherent, and*
(2) *for every flat, locally finitely presented morphism $g : U \to S$ the canonical map $g^*_{\tau,small}(\mathcal{G}|_{S_\tau}) \to \mathcal{G}|_{U_\tau}$ is an isomorphism.*

Then $H^p(U, \mathcal{G}) = H^p(U, \mathcal{G}|_{U_\tau})$ for every U flat and locally of finite presentation over S.

Proof. Let $\mathcal{F}$ be the pullback of $\mathcal{G}|_{S_\tau}$ to the big fppf site $(Sch/S)_{fppf}$. Note that $\mathcal{F}$ is quasi-coherent. There is a canonical comparison map $\varphi : \mathcal{F} \to \mathcal{G}$ which by assumptions (1) and (2) induces an isomorphism $\mathcal{F}|_{U_\tau} \to \mathcal{G}|_{U_\tau}$ for all $g : U \to S$ flat and locally of finite presentation. Hence in the short exact sequences

$$0 \to \mathrm{Ker}(\varphi) \to \mathcal{F} \to \mathrm{Im}(\varphi) \to 0$$

and

$$0 \to \mathrm{Im}(\varphi) \to \mathcal{G} \to \mathrm{Coker}(\varphi) \to 0$$

the sheaves $\mathrm{Ker}(\varphi)$ and $\mathrm{Coker}(\varphi)$ are parasitic for the fppf topology. By Lemma 0 we conclude that $H^p(U, \mathcal{F}) \to H^p(U, \mathcal{G})$ is an isomorphism for $g : U \to S$ flat and locally of finite presentation. Since the result holds for $\mathcal{F}$ by Proposition 0 we win. □